

Crowd-Sourced Civic Issue Reporting & Resolution System

A complete full-stack project structure using:

- React (Frontend)
- CSS (Responsive UI)
- Node.js + Express.js (Backend)
- MongoDB Atlas (Database)
- fetch() API calls
- Frontend and backend deployed separately

1. Project Structure

```
civic-issue-system/  
|  
├─ client/    (React Frontend)  
|  
└─ server/    (Express Backend)
```

2. Backend Setup (server)

Install

```
mkdir server  
cd server  
npm init -y  
npm install express mongoose cors dotenv
```

Folder Structure

```
server/  
|  
├─ models/  
|   ├── User.js  
|   └── Issue.js
```

```
|
|├─ routes/
|   │├─ userRoutes.js
|   └─ issueRoutes.js
|
|├─ .env
└─ server.js
```

.env

```
MONGO_URI=your_mongodb_connection_string
PORT=5000
```

server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
require("dotenv").config();

const app = express();

app.use(cors());
app.use(express.json());

const userRoutes = require("./routes/userRoutes");
const issueRoutes = require("./routes/issueRoutes");

app.use("/api/users", userRoutes);
app.use("/api/issues", issueRoutes);

mongoose.connect(process.env.MONGO_URI)
  .then(() => console.log("MongoDB Connected"))
  .catch(err => console.log(err));

app.listen(process.env.PORT, () => {
  console.log("Server running on port 5000");
});
```

models/User.js

```
const mongoose = require("mongoose");

const userSchema = new mongoose.Schema({
  username: String,
  email: String,
  password: String,
  role: {
    type: String,
    default: "user"
  }
});

module.exports = mongoose.model("User", userSchema);
```

models/Issue.js

```
const mongoose = require("mongoose");

const issueSchema = new mongoose.Schema({
  title: String,
  description: String,
  location: String,
  status: {
    type: String,
    default: "Pending"
  },
  username: String
}, { timestamps: true });

module.exports = mongoose.model("Issue", issueSchema);
```

routes/userRoutes.js

```
const express = require("express");
const router = express.Router();
const User = require("../models/User");

router.post("/register", async (req, res) => {
  const user = new User(req.body);
  await user.save();
  res.json(user);
});
```

```

});

router.post("/login", async (req, res) => {
  const { email, password } = req.body;

  const user = await User.findOne({
    email,
    password
  });

  if (!user) {
    return res.status(400).json({
      message: "Invalid credentials"
    });
  }

  res.json(user);
});

module.exports = router;

```

routes/issueRoutes.js

```

const express = require("express");
const router = express.Router();
const Issue = require("../models/Issue");

router.post("/", async (req, res) => {
  const issue = new Issue(req.body);
  await issue.save();
  res.json(issue);
});

router.get("/", async (req, res) => {
  const issues = await Issue.find();
  res.json(issues);
});

router.put("/:id", async (req, res) => {
  const updated = await Issue.findByIdAndUpdate(
    req.params.id,
    req.body,
    { new: true }
  );

  res.json(updated);
});

```

```
});  
  
module.exports = router;
```

Run backend:

```
node server.js
```

3. Frontend Setup (client)

Install

```
npx create-react-app client  
cd client  
npm install react-router-dom
```

Structure

```
src/  
|  
├── components/  
│   └── Navbar.js  
|  
├── pages/  
│   ├── Home.js  
│   ├── Login.js  
│   ├── Register.js  
│   ├── Dashboard.js  
│   ├── ReportIssue.js  
│   └── AdminDashboard.js  
|  
├── App.js  
└── App.css
```

Navbar.js

```
import './Navbar.css';  
  
function Navbar() {
```

```

const user = JSON.parse(localStorage.getItem("user"));

return (
  <nav className="navbar">
    <div className="left-section">
      <div className="logo"><img alt="Globe icon" data-bbox="408 178 431 191"/></div>
      <h2>CivicConnect</h2>
      <div className="hamburger"><img alt="Hamburger menu icon" data-bbox="451 211 474 224"/></div>
    </div>

    <div className="right-section">
      <span><img alt="User profile icon" data-bbox="268 278 291 291"/></span>
      <span>
        {user?.role === "admin"
          ? "Admin"
          : user?.username || "Guest"}
      </span>
    </div>
  </nav>
);
}

export default Navbar;

```

Navbar.css

```

.navbar {
  width: 100%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: #0f172a;
  color: white;
  padding: 15px 20px;
  box-sizing: border-box;
}

.left-section {
  display: flex;
  align-items: center;
  gap: 12px;
}

.hamburger {
  cursor: pointer;
}

```

```

.right-section {
  display: flex;
  align-items: center;
  gap: 8px;
}

@media (max-width: 768px) {
  .navbar {
    padding: 12px;
  }
}

```

Login.js

```

import { useState } from "react";

function Login() {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");

  const handleLogin = async () => {
    const res = await fetch(
      "http://localhost:5000/api/users/login",
      {
        method: "POST",
        headers: {
          "Content-Type": "application/json"
        },
        body: JSON.stringify({ email, password })
      }
    );

    const data = await res.json();

    localStorage.setItem(
      "user",
      JSON.stringify(data)
    );

    alert("Login Success");
  };

  return (
    <div>
      <h2>Login</h2>

```

```

    <input
      placeholder="Email"
      onChange={(e) => setEmail(e.target.value)}
    />

    <input
      placeholder="Password"
      onChange={(e) => setPassword(e.target.value)}
    />

    <button onClick={handleLogin}>
      Login
    </button>
  </div>
);
}

export default Login;

```

Register.js

```

import { useState } from "react";

function Register() {
  const [formData, setFormData] = useState({
    username: "",
    email: "",
    password: ""
  });

  const handleRegister = async () => {
    await fetch(
      "http://localhost:5000/api/users/register",
      {
        method: "POST",
        headers: {
          "Content-Type": "application/json"
        },
        body: JSON.stringify(formData)
      }
    );

    alert("Registered Successfully");
  };
}

```



```

return (
  <div>
    <input
      placeholder="Username"
      onChange={e =>
        setFormData({
          ...formData,
          username: e.target.value
        })
      }
    />

    <input
      placeholder="Email"
      onChange={e =>
        setFormData({
          ...formData,
          email: e.target.value
        })
      }
    />

    <input
      placeholder="Password"
      onChange={e =>
        setFormData({
          ...formData,
          password: e.target.value
        })
      }
    />

    <button onClick={handleRegister}>
      Register
    </button>
  </div>
);
}

export default Register;

```

ReportIssue.js

```

import { useState } from "react";

function ReportIssue() {

```

```

const [title, setTitle] = useState("");
const [description, setDescription] = useState("");
const [location, setLocation] = useState("");

const submitIssue = async () => {
  const user = JSON.parse(localStorage.getItem("user"));

  await fetch(
    "http://localhost:5000/api/issues",
    {
      method: "POST",
      headers: {
        "Content-Type": "application/json"
      },
      body: JSON.stringify({
        title,
        description,
        location,
        username: user.username
      })
    }
  );

  alert("Issue Submitted");
};

return (
  <div>
    <h2>Report Issue</h2>
    <input placeholder="Title" onChange={(e)=>setTitle(e.target.value)} />
    <input placeholder="Description"
  onChange={(e)=>setDescription(e.target.value)} />
    <input placeholder="Location"
  onChange={(e)=>setLocation(e.target.value)} />
    <button onClick={submitIssue}>Submit</button>
  </div>
);
}

export default ReportIssue;

```

AdminDashboard.js

```

import { useEffect, useState } from "react";

function AdminDashboard() {

```

```

const [issues, setIssues] = useState([]);

useEffect(() => {
  fetch("http://localhost:5000/api/issues")
    .then(res => res.json())
    .then(data => setIssues(data));
}, []);

return (
  <div>
    <h2>All Issues</h2>

    {issues.map(issue => (
      <div key={issue._id}>
        <h3>{issue.title}</h3>
        <p>{issue.description}</p>
        <p>{issue.status}</p>
      </div>
    ))}
  </div>
);
}

export default AdminDashboard;

```

App.js

```

import { BrowserRouter, Routes, Route } from "react-router-dom";
import Navbar from "./components/Navbar";
import Login from "./pages/Login";
import Register from "./pages/Register";
import ReportIssue from "./pages/ReportIssue";
import AdminDashboard from "./pages/AdminDashboard";

function App() {
  return (
    <BrowserRouter>
      <Navbar />

      <Routes>
        <Route path="/" element={<Login />} />
        <Route path="/register" element={<Register />} />
        <Route path="/report" element={<ReportIssue />} />
        <Route path="/admin" element={<AdminDashboard />} />
      </Routes>
    </BrowserRouter>
  );
}

```

```
    );  
  }  
  
  export default App;
```

4. Deployment

Backend:

- Deploy server on Render

Frontend:

- Deploy React app on Vercel

Replace localhost URL with deployed backend URL after deployment.